Precept 6 Outline

ECO 202 TAs

October 15, 2026

1 Weekly Check In

1. Exam 2 grades released!
2. Any questions about the class, comments?
3. Reminders about resources:
   - Come to office hours for any preceptor, as much as you want.
     - Dexin: Tuesdays, 4–5pm, tables outside of JRR A19
     - Emily: Thursdays, 4:30–5:30pm, JRR A74
     - Ryan: Fridays, 2–3pm, JRR A74
   - Come to office hours of the professor: Fridays, 3pm–4pm, JRR 278
   - Check out the McGraw Center:
     - Individual tutoring: Link to page
     - Group study with a tutor: Link to page
   - Reach out to us by email if you've tried the above but are seeking support beyond these things.

2 Precept practice problems

1. **(Practice 10, Question 4)**

   For a stylized delivery business, let R be the next-day revenue change, measured in hundreds of dollars, with possible values 0 and 2. Let $Q = 1$ if all scheduled deliveries arrive on time and $Q = 0$ otherwise. Their joint distribution is

Delivery indicator	$R = 0$	$R = 2$	Marginal probability
$Q = 0$	0.30	0.10	
$Q = 1$	0.20	0.40	
Marginal probability			1.00

   (a) Complete all marginal probabilities and verify that the four joint cells form a valid distribution.

   Solution. Summing rows and columns gives

Delivery indicator	$R = 0$	$R = 2$	Marginal probability
$Q = 0$	0.30	0.10	0.40
$Q = 1$	0.20	0.40	0.60
Marginal probability	0.50	0.50	1.00

   All cells are nonnegative, and they sum to one.

(b) Write the marginal pmfs of R and Q, then calculate $\mathbb{E}[R]$ and $\mathbb{E}[Q]$.

Solution. The marginal pmfs are

$$\mathbb{P}(R=0)=0.50, \qquad \mathbb{P}(R=2)=0.50$$

and

$$\mathbb{P}(Q=0)=0.40, \qquad \mathbb{P}(Q=1)=0.60$$

Therefore

$$\mathbb{E}[R]=0(0.50)+2(0.50)=1$$

hundred dollars, and

$$\mathbb{E}[Q]=0(0.40)+1(0.60)=0.60$$

(c) Calculate $\mathrm{Var}(R)$, $\mathrm{SD}(R)$, $\mathrm{Var}(Q)$, and $\mathrm{SD}(Q)$.

Solution. For revenue,

$$\mathbb{E}[R^2]=0^2(0.50)+2^2(0.50)=2$$

so

$$\mathrm{Var}(R)=2-1^2=1, \qquad \mathrm{SD}(R)=1$$

For the indicator,

$$\mathrm{Var}(Q)=0.60(0.40)=0.24, \qquad \mathrm{SD}(Q)=\sqrt{0.24}\approx 0.4899$$

(d) Calculate $\mathbb{E}[RQ]$ directly from the joint table.

Solution. Only the cell $(R=2, Q=1)$ contributes a nonzero product, so

$$\mathbb{E}[RQ]=2(1)(0.40)=0.80$$

(e) Calculate $\mathrm{Cov}(R,Q)$ and $\mathrm{Corr}(R,Q)$. State the units of each.

Solution. We have

$$\mathrm{Cov}(R,Q)=\mathbb{E}[RQ]-\mathbb{E}[R]\mathbb{E}[Q]=0.80-(1)(0.60)=0.20$$

and

$$\mathrm{Corr}(R,Q)=\frac{0.20}{(1)(\sqrt{0.24})}\approx 0.4082$$

Covariance has units of hundreds of dollars times the unitless indicator, while correlation is unitless.

(f) Test independence using the cell $(R=2, Q=1)$. Would the sign of the covariance alone be enough to establish independence or causality? Explain.

Solution. Independence fails because

$$\mathbb{P}(R=2,Q=1)=0.40\neq \mathbb{P}(R=2)\mathbb{P}(Q=1)=(0.50)(0.60)=0.30$$

A nonzero covariance also rules out independence, but zero covariance would not establish independence in general. Neither the positive covariance nor the failed factorization establishes causality; no assignment or other causal design is supplied.

2. **(Practice 11, Question 2)**

A two-record teaching population contains payment values 3 and 9, measured in tens of dollars. Select two records independently and uniformly with replacement and calculate $\overline{Y}_2$.

(a) List all four equally likely ordered samples and calculate $\overline{Y}_2$ for each.

Solution. The four ordered samples and their means are

Ordered sample	$(3,3)$	$(3,9)$	$(9,3)$	$(9,9)$
Sample mean	3	6	6	9

(b) Construct the sampling distribution of $\overline{Y}_2$ and verify that its probabilities sum to one.

Solution. The sampling distribution is

$\overline{y}_2$	3	6	9
Probability	1/4	1/2	1/4

The probabilities are nonnegative and sum to one.

(c) Calculate $\mathbb{E}[\overline{Y}_2]$, $\mathrm{Var}(\overline{Y}_2)$, and $\mathrm{SD}(\overline{Y}_2)$ directly from that distribution.

Solution. Direct calculation gives

$$\mathbb{E}[\overline{Y}_2] = 3\left(\frac{1}{4}\right) + 6\left(\frac{1}{2}\right) + 9\left(\frac{1}{4}\right) = 6$$

and

$$\mathbb{E}[\overline{Y}_2^2] = 9\left(\frac{1}{4}\right) + 36\left(\frac{1}{2}\right) + 81\left(\frac{1}{4}\right) = 40.5$$

Hence

$$\mathrm{Var}(\overline{Y}_2) = 40.5 - 6^2 = 4.5 = \frac{9}{2}$$

and

$$\mathrm{SD}(\overline{Y}_2) = \sqrt{\frac{9}{2}} = \frac{3}{\sqrt{2}} \approx 2.1213$$

in tens of dollars.

(d) Calculate the population mean μ_Y and population variance σ_Y^2 of one uniform draw. Verify that $\mathbb{E}[\overline{Y}_2] = \mu_Y$ and $\mathrm{Var}(\overline{Y}_2) = \sigma_Y^2/2$.

Solution. For one uniform population draw,

$$\mu_Y = \frac{3+9}{2} = 6$$

and

$$\sigma_Y^2 = \frac{(3-6)^2 + (9-6)^2}{2} = 9$$

Therefore

$$\mathbb{E}[\overline{Y}_2] = \mu_Y, \qquad \mathrm{Var}(\overline{Y}_2) = \frac{9}{2} = \frac{\sigma_Y^2}{2}$$

(e) If the realized ordered sample is $(3, 9)$, identify the realized sample distribution and realized sample mean. Explain which full sampling-distribution quantities do not change after observing it.

Solution. The realized sample distribution places one of its two observed values at 3 and one at 9, and the realized sample mean is 6. Observing the sample does not change the population distribution, the complete sampling distribution, its expectation, or its variance.

3. **(Practice 11, Question 3)**

Under an explicitly stated independent with-replacement sampling model, let $H_i = 1$ when draw i has a specified characteristic and $H_i = 0$ otherwise, with $\mathbb{P}(H_i = 1) = p = 0.36$. Define

$$C_n = \sum_{i=1}^{n} H_i, \qquad \widehat{P}_n = \frac{C_n}{n}$$

(a) Starting from the indicator mean and variance, derive $\mathbb{E}[C_n]$, $\mathrm{SD}(C_n)$, $\mathbb{E}[\widehat{P}_n]$, and $\mathrm{SE}(\widehat{P}_n)$.

Solution. For each indicator,

$$\mathbb{E}[H_i] = p = 0.36, \qquad \mathrm{Var}(H_i) = p(1-p) = (0.36)(0.64) = 0.2304$$

Linearity of expectation gives

$$\mathbb{E}[C_n] = np$$

Independence makes the covariance terms zero, so

$$\mathrm{Var}(C_n) = np(1-p), \qquad \mathrm{SD}(C_n) = \sqrt{np(1-p)}$$

Because $\widehat{P}_n = C_n/n$,

$$\mathbb{E}[\widehat{P}_n] = p, \qquad \mathrm{SE}(\widehat{P}_n) = \sqrt{\frac{p(1-p)}{n}}$$

(b) Calculate all four quantities for $n = 25$.

Solution. For $n = 25$,

$$\mathbb{E}[C_{25}] = 25(0.36) = 9, \qquad \mathrm{SD}(C_{25}) = \sqrt{25(0.2304)} = 2.4$$

and

$$\mathbb{E}[\widehat{P}_{25}] = 0.36, \qquad \mathrm{SE}(\widehat{P}_{25}) = \sqrt{\frac{0.2304}{25}} = 0.096$$

(c) Calculate all four quantities for $n = 100$.

Solution. For $n = 100$,

$$\mathbb{E}[C_{100}] = 36, \qquad \mathrm{SD}(C_{100}) = \sqrt{100(0.2304)} = 4.8$$

and

$$\mathbb{E}[\widehat{P}_{100}] = 0.36, \qquad \mathrm{SE}(\widehat{P}_{100}) = \sqrt{\frac{0.2304}{100}} = 0.048$$

(d) Explain why every realized count is an integer although an expected count need not be.

Solution. A realized count is a sum of zeros and ones and must be an integer. Its expectation is a probability-weighted center over possible counts and need not itself be a possible count.

(e) When n is multiplied by four, state the exact factor change in the count center, count standard deviation, proportion center, and proportion standard error.

Solution. The count center np is multiplied by four, the count standard deviation $\sqrt{np(1-p)}$ is multiplied by two, the proportion center p is unchanged, and the proportion standard error is divided by two.

(f) Identify which steps require independence and which expectation steps use linearity without requiring independence.

Solution. Independence is required to add variances without covariance terms. Linearity supplies both expectation formulas without requiring independence, provided the marginal indicator probabilities remain p.

3 Extra practice problems

1. The random variables X and Y have marginal distributions

x	1	2
$\mathbb{P}(X = x)$	0.40	0.60

y	3	4
$\mathbb{P}(Y = y)$	0.30	0.70

(a) Calculate $\mathrm{Var}(X)$.

Solution. We have

$$\mathbb{E}[X] = 0.40(1) + 0.60(2) = 1.6$$

and

$$\mathbb{E}[X^2] = 0.40(1^2) + 0.60(2^2) = 2.8$$

Therefore

$$\mathrm{Var}(X) = 2.8 - (1.6)^2 = 0.24$$

(b) Suppose X and Y are independent. Find the probability distribution of $W = Y - X$.

Solution. Independence gives

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y)$$

Hence

$$\mathbb{P}(W = 1) = \mathbb{P}(X = 2, Y = 3) = (0.60)(0.30) = 0.18$$
$$\mathbb{P}(W = 2) = \mathbb{P}(X = 1, Y = 3) + \mathbb{P}(X = 2, Y = 4) = (0.40)(0.30) + (0.60)(0.70) = 0.54$$

and

$$\mathbb{P}(W = 3) = \mathbb{P}(X = 1, Y = 4) = (0.40)(0.70) = 0.28$$

Thus

w	1	2	3
$\mathbb{P}(W = w)$	0.18	0.54	0.28

(c) Suppose X and Y are not necessarily independent but retain the marginal distributions above. What is the largest possible value of $\mathbb{E}[W]$? Explain.

Solution. Linearity of expectation does not require independence. Therefore

$$\mathbb{E}[W] = \mathbb{E}[Y - X] = \mathbb{E}[Y] - \mathbb{E}[X]$$

The marginals determine

$$\mathbb{E}[Y] = 3(0.30) + 4(0.70) = 3.7$$

and $\mathbb{E}[X] = 1.6$, so every possible dependence structure gives

$$\mathbb{E}[W] = 3.7 - 1.6 = \boxed{2.1}$$

Thus the largest possible value is also 2.1.

2. For working-age males in the United States, suppose the number of jobs held at the same time has distribution

Number of jobs	0	1	2
Probability	0.20	0.75	0.05

For working-age females, suppose the distribution is

Number of jobs	0	1
Probability	0.50	0.50

Assume there are equally many working-age males and females.

(a) Randomly select one male and one female and calculate their average number of jobs. Find the probability distribution of this average.

Solution. Let X be the male's number of jobs, let Y be the female's number of jobs, and let $Z = (X+Y)/2$. The two selections are independent. The possible values and probabilities are

$$\mathbb{P}(Z = 0) = (0.20)(0.50) = 0.10$$

$$\mathbb{P}(Z = 0.5) = (0.20)(0.50) + (0.75)(0.50) = 0.475$$

$$\mathbb{P}(Z = 1) = (0.75)(0.50) + (0.05)(0.50) = 0.40$$

and

$$\mathbb{P}(Z = 1.5) = (0.05)(0.50) = 0.025$$

Therefore

z	0	0.5	1	1.5
$\mathbb{P}(Z = z)$	0.10	0.475	0.40	0.025

(b) Randomly select one individual from all working-age people and ask for that person's number of jobs. Find its probability distribution.

Solution. Let J be the selected person's number of jobs. Because half the population is male and half is female,

$$\mathbb{P}(J = 0) = \frac{1}{2}(0.20) + \frac{1}{2}(0.50) = 0.35$$

$$\mathbb{P}(J = 1) = \frac{1}{2}(0.75) + \frac{1}{2}(0.50) = 0.625$$

and

$$\mathbb{P}(J = 2) = \frac{1}{2}(0.05) + \frac{1}{2}(0) = 0.025$$

Thus

j	0	1	2
$\mathbb{P}(J = j)$	0.35	0.625	0.025

3. Suppose X has distribution

x	-1	0	1
$\mathbb{P}(X = x)$	0.30	0.20	0.50

Let $X_1, X_2, \ldots$ be independent random variables with the same distribution as X, and define

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$$

(a) Compute $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

Solution. The mean is

$$\mathbb{E}[X] = (-1)(0.30) + 0(0.20) + 1(0.50) = 0.20$$

and

$$\mathrm{Var}(X) = (-1 - 0.20)^2(0.30) + (0 - 0.20)^2(0.20) + (1 - 0.20)^2(0.50) = 0.76$$

(b) What is the probability that $\overline{X}_2 > 0$?

Solution. The only ordered pairs that yield a positive mean are $(0,1)$, $(1,0)$, and $(1,1)$. By independence,

$$\mathbb{P}(0,1) = (0.20)(0.50) = 0.10, \qquad \mathbb{P}(1,0) = 0.10$$

and

$$\mathbb{P}(1,1) = (0.50)(0.50) = 0.25$$

Therefore

$$\mathbb{P}(\overline{X}_2 > 0) = 0.10 + 0.10 + 0.25 = \boxed{0.45}$$

(c) What values does $\overline{X}_{10}$ take with positive probability?

Solution. The sum $X_1 + \cdots + X_{10}$ can equal every integer from -10 through 10. Therefore

$$\overline{X}_{10} \in \left\{ \frac{k}{10} : k = -10, -9, \ldots, 9, 10 \right\}$$

(d) Explain why $\sum_{i=1}^{n} X_i^2$ is exactly binomial, and give its parameters.

Solution. Each X_i^2 equals zero with probability 0.20 and one with probability 0.80. The variables X_i^2 are independent Bernoulli variables with success probability 0.80. Hence

$$\sum_{i=1}^{n} X_i^2 \sim \text{Binomial}(n, 0.80)$$

4. Let X and Y be random variables satisfying

$$\text{Var}(X) = 3, \qquad \text{Var}(Y) = 10, \qquad \text{Var}(X+Y) = 9$$

What is the correlation between X and Y?

Solution. Using the variance-of-a-sum formula,

$$9 = 3 + 10 + 2\,\text{Cov}(X,Y)$$

so

$$\text{Cov}(X,Y) = -2$$

Therefore

$$\text{Corr}(X,Y) = \frac{-2}{\sqrt{3}\sqrt{10}} = \frac{-2}{\sqrt{30}} \approx \boxed{-0.365}$$

5. The random variable X has distribution

x	-2	4	5
$\mathbb{P}(X = x)$	0.30	0.50	0.20

(a) Compute the mean and variance of X.

Solution. The mean is

$$\mathbb{E}[X] = (-2)(0.30) + 4(0.50) + 5(0.20) = 2.4$$

and

$$\text{Var}(X) = (-2-2.4)^2(0.30) + (4-2.4)^2(0.50) + (5-2.4)^2(0.20) = 8.44$$

(b) Let X_1 and X_2 be independent random variables with the same distribution as X. What is $\mathbb{P}(X_1 + X_2 < 0)$? Explain.

Solution. The sum is negative only when both variables equal -2. By independence,

$$\mathbb{P}(X_1 + X_2 < 0) = \mathbb{P}(X_1 = -2, X_2 = -2) = (0.30)(0.30) = \boxed{0.09}$$

(c) A random variable Y satisfies $\mathbb{E}[X+Y] = 0$ and $\text{Var}(X+Y) = 0$. What is the joint distribution of X and Y?

Solution. Let $Z = X + Y$. Since $\text{Var}(Z) = 0$, the variable Z equals its mean with probability one. Because $\mathbb{E}[Z] = 0$, we must have $Z = 0$ with probability one, so $Y = -X$. Therefore

(x,y)	$(-2,2)$	$(4,-4)$	$(5,-5)$
$\mathbb{P}(X = x, Y = y)$	0.30	0.50	0.20

6. Suppose (Y, X) has joint distribution

	$X=-1$	$X=1$	$X=2$
$Y=2$	0.10	0.35	0.20
$Y=3$	0.05	0.20	0.10

(a) Are X and Y independent? Explain.

Solution. The relevant marginals are $\mathbb{P}(X=-1)=0.15$ and $\mathbb{P}(Y=2)=0.65$. Since

$$\mathbb{P}(X=-1, Y=2)=0.10 \neq (0.15)(0.65)=0.0975$$

the variables are not independent.

(b) What is $\mathrm{Var}(X)$? Explain.

Solution. The marginal distribution is

$$\mathbb{P}(X=-1)=0.15, \qquad \mathbb{P}(X=1)=0.55, \qquad \mathbb{P}(X=2)=0.30$$

Thus

$$\mathbb{E}[X]=(-1)(0.15)+1(0.55)+2(0.30)=1$$

and

$$\mathrm{Var}(X)=(-1-1)^2(0.15)+(1-1)^2(0.55)+(2-1)^2(0.30)=0.90$$

(c) What is $\mathbb{P}(Y>X)$? Explain.

Solution. The event includes every cell in the $Y=3$ row, together with $(X,Y)=(-1,2)$ and $(1,2)$. Therefore

$$\mathbb{P}(Y>X)=0.35+0.10+0.35=\boxed{0.80}$$

(d) Find the distribution of $W=X^2-Y$.

Solution. Mapping each joint outcome into $W=X^2-Y$ gives

(x,y)	$(-1,2)$	$(1,2)$	$(2,2)$	$(-1,3)$	$(1,3)$	$(2,3)$
Probability	0.10	0.35	0.20	0.05	0.20	0.10
$w=x^2-y$	-1	-1	2	-2	-2	1

Combining equal values gives

w	-2	-1	1	2
$\mathbb{P}(W=w)$	0.25	0.45	0.10	0.20

7. A population consists of three individuals A, B, and C, whose values are -2, 0, and 4, respectively. A simple random sample of size two is taken without replacement. Find the sampling distribution of the sample mean and the median of that sampling distribution.

Solution. The three equally likely samples are

$$\{A,B\}, \qquad \{A,C\}, \qquad \{B,C\}$$

Their sample means are

$$\frac{-2+0}{2}=-1, \qquad \frac{-2+4}{2}=1, \qquad \frac{0+4}{2}=2$$

Thus

$\overline{x}$	-1	1	2
$\mathbb{P}(\overline{X}=\overline{x})$	1/3	1/3	1/3

The median of the sampling distribution is $\boxed{1}$. The official solution contains an arithmetic typo in the third mean; $(0+4)/2$ is 2, not -2.

8. In a population of four individuals, a quantitative variable takes the values

$$\{1,3,4,5\}$$

A simple random sample of size three is taken. Find the sampling distribution of the sample median.

Solution. The four equally likely samples are

$$\{1,3,4\}, \quad \{1,3,5\}, \quad \{1,4,5\}, \quad \{3,4,5\}$$

The first two have median 3, and the last two have median 4. Therefore

m	3	4
$\mathbb{P}(M = m)$	0.50	0.50

9. Suppose x is a quantitative variable for a set of N individuals. Consider the random phenomenon of selecting one individual uniformly at random, and let X be the selected individual's value of x.

   (a) Describe the distribution of X in terms of the values $x_1, \ldots, x_N$.

   Solution. For any number a appearing in the population,

   $$\mathbb{P}(X = a) = \frac{\#\{i : x_i = a\}}{N}$$

   Equivalently, each indexed individual i is selected with probability $1/N$, so repeated values have their probabilities combined.

   (b) What is $\mathbb{E}[X]$? Does it matter whether several individuals have the same value?

   Solution. The expectation is the population mean

   $$\mathbb{E}[X] = \frac{1}{N}\sum_{i=1}^{N} x_i = \mu$$

   Ties do not change the formula because each individual contributes once to the sum.

   (c) What is $\mathrm{Var}(X)$?

   Solution. The variance is the population variance with denominator N:

   $$\mathrm{Var}(X) = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2 = \sigma^2$$

   (d) Now select n individuals independently with replacement. Let $X_1, \ldots, X_n$ be their values and let

   $$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

   Find $\mathbb{E}[\overline{X}]$ and $\mathrm{Var}(\overline{X})$.

   Solution. Linearity of expectation gives

   $$\mathbb{E}[\overline{X}] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] = \frac{1}{n}(n\mu) = \mu$$

   Because sampling is independent with replacement,

   $$\mathrm{Var}(\overline{X}) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n}$$

   Consequently,

   $$\mathrm{SD}(\overline{X}) = \frac{\sigma}{\sqrt{n}}$$